

CS2227 | AI & ML

Module 1: Introduction to Artificial Intelligence

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| Learning Objectives



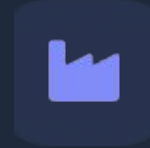
History

Trace the evolution of AI from 1943 to 2025.



Levels

Distinguish between levels of AI (ANI, AGI, ASI).



Impact

Connect AI concepts to real-world industries.



Algorithms

Execute and compare Uninformed vs. Informed search.

| Defining Artificial Intelligence

Standard Definition

The field of study that seeks to explain and emulate intelligent behavior in terms of computational processes.

Core Question: Can machines think, or just act as if they think? ([The Turing Test](#)).

Mitchell's Perspective

Focuses on "**Learning**".

A program improves performance **P** at a task **T** through experience **E**.

"Machine learning is the study of computer algorithms that allow computer programs to automatically improve through experience."

The First Wave: Cybernetics (1940s–60s)

1943: McCulloch-Pitts Neuron

First mathematical model of a biological neuron using binary logic gates. Inspired by biological neurons.

1958: The Perceptron

Frank Rosenblatt's algorithm that could learn to classify data points. The ancestor of modern neural nets.



The Limitation

Could only solve linearly separable problems.

Famous Failure: The "XOR" Problem

| The First AI Winter (1970s)

- **The Minsky-Papert Critique:** Rigorously proved that Perceptrons were strictly limited in what they could compute.
- **The Fallout:** Funding for neural network research essentially dried up overnight.
- **The Shift:** Research focus moved almost entirely to symbolic AI and rigid "if-then" logic systems.

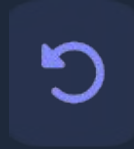


| The Second Wave: Connectionism (1980s–90s)



Philosophy

Knowledge is not stored in symbols, but distributed across a massive network of connections.



1986: Backpropagation

Rumelhart, Hinton, & Williams popularized the algorithm allowing multi-layer networks to learn from errors.



1998: CNNs (LeNet-5)

Yann LeCun developed CNNs for digit recognition (zip codes), creating the foundation for computer vision.

The Third Wave: Deep Learning (2006–12)

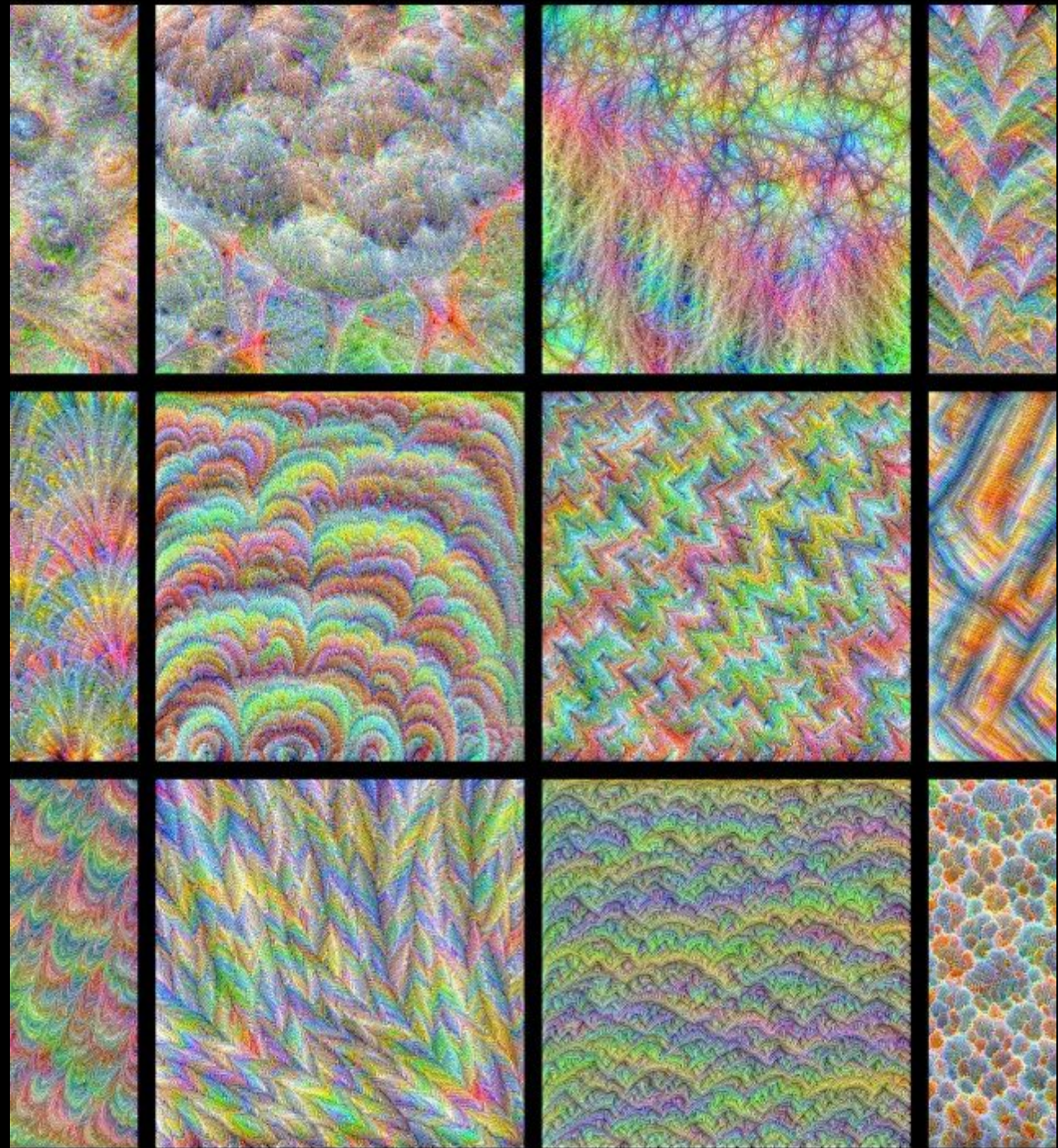
The Catalyst: Explosion of Big Data (ImageNet) and GPU computing.

2006: Pre-training

Geoffrey Hinton's breakthrough allowing training of very deep networks without vanishing gradients.

2012: AlexNet

A deep CNN shattered records in the ImageNet challenge. The start of the modern AI era.



| The Current Era: Transformers (2017–24)



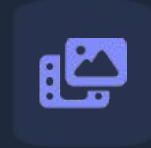
2017: The Transformer

Paper: "Attention is All You Need". Allowed models to process whole sequences of data at once, replacing RNNs.



LLMs

Scaling Transformers led to GPT-3/4, Claude, and Gemini. Emergent capabilities in language.



Multi-modal AI

Models that natively understand and generate text, audio, and video simultaneously.

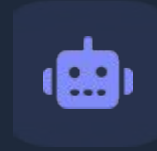
Modern Trends: GenAI & Agents



Generative AI

Moving from "recognizing" patterns to "creating" new content.

- Images (Midjourney)
- Video (Sora)
- Code (Copilot)



AI Agents

Models that don't just chat, but take action.

- Use tools (Browsers, Python)
- Complete multi-step tasks autonomously
- "Agentic" workflows

| Type 1: Artificial Narrow Intelligence (ANI)

- **Scope:** Strictly task-specific.
- **Capabilities:** Excels at one thing, often better than humans.
- **Examples:**
 - Playing Chess (Deep Blue)
 - Diagnosing skin cancer
 - Spotify recommendations
- **Status:** This is our current reality.



| Type 2: Artificial General Intelligence (AGI)

- **Scope:** Human-level adaptability.
- **Capabilities:** A single system that can learn to:
 - Paint art
 - Write code
 - Solve physics equations
 - Exhibit social empathy
- **Status:** Under active research. Highly debated timeline (5 vs 50 years).



| Type 3: Artificial Super Intelligence (ASI)

- **Scope:** Surpassing all human cognitive abilities.
- **Capabilities:** Problem-solving speeds, memory, and creativity far beyond the limits of biological brains.
- **Context:** Often discussed in AI Safety and Ethics.
- **Status:** Theoretical.



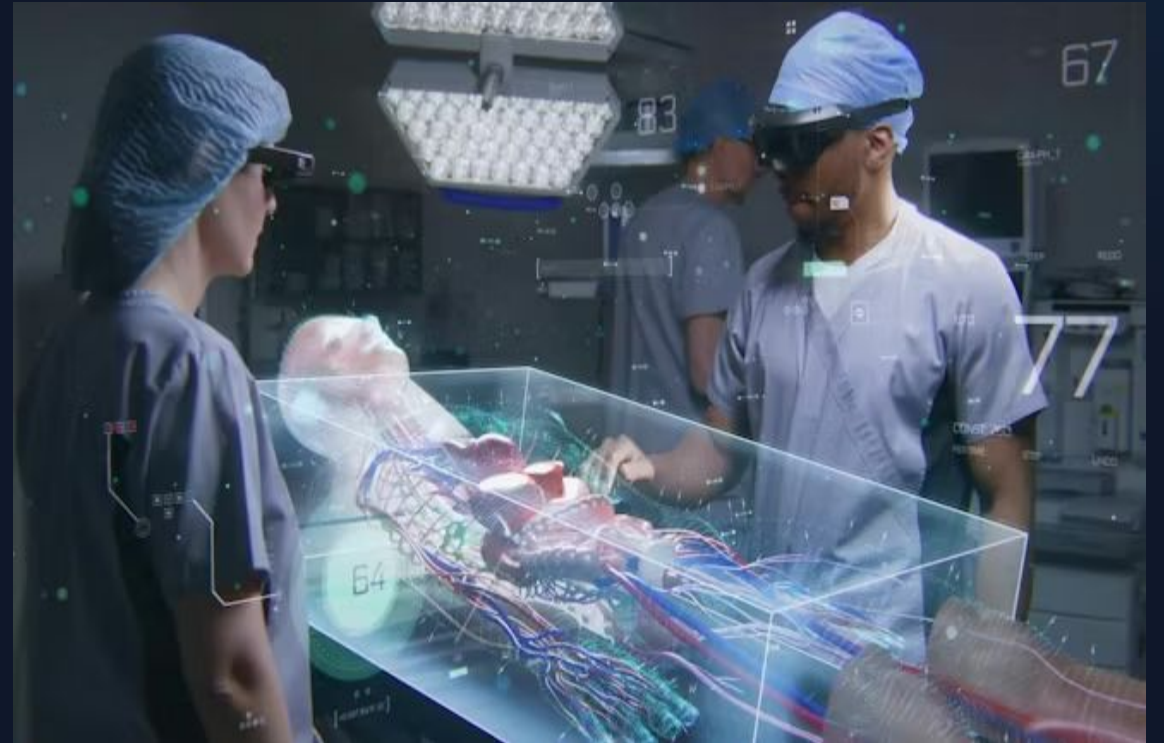
| Domain: Healthcare

AI in Pathology

Tech: CNNs (Convolutional Neural Nets) analyze biopsies faster and more accurately than human eyes.

Benefit

Early detection of tumors and accelerated personalized drug discovery.



Domain: Finance

HFT & Fraud Detection

Tech: RNNs (Recurrent Neural Networks) analyze time-series data to predict market shifts in milliseconds.

Benefit

Instantaneous detection of unauthorized credit card transactions.



| Domain: Robotics

Warehouse Automation

Tech: Reinforcement Learning (RL) allows robots to learn how to grasp objects of different shapes and sizes through trial and error.

Benefit

24/7 logistics efficiency (e.g., Amazon Robotics).



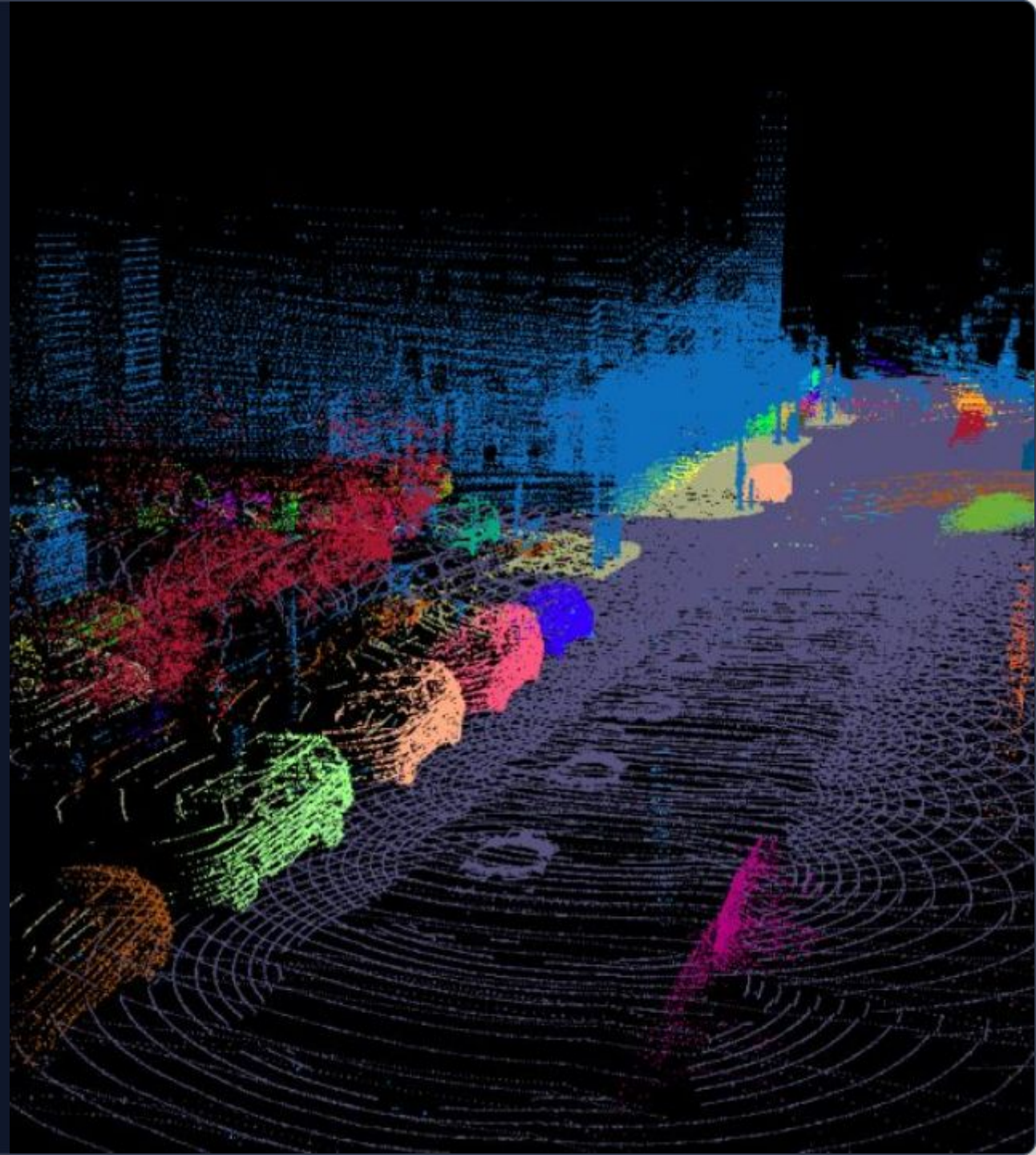
Domain: Autonomous Vehicles

Tesla / Waymo

Tech: Sensor Fusion. Combining Camera, Lidar, and Radar data processed by Deep Neural Networks.

Benefit

Reducing human-error-related traffic accidents.



| Part 2: Problem Solving as Search



State Space

All possible configurations of a problem (the "map").



Operators

Actions that move from one state to another.



Goal Test

Does the current state satisfy the solution?

| Uninformed Search: BFS

Breadth-First Search

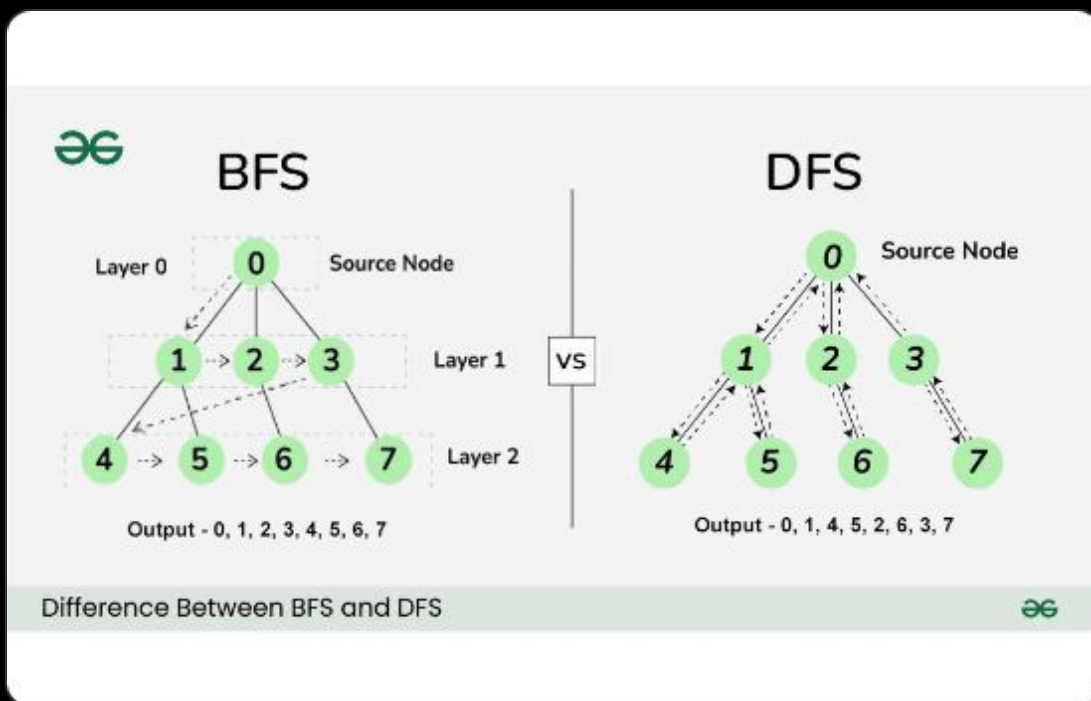
Logic: Explore the "frontier" layer by layer.

 Mechanism: **Queue** (First-In, First-Out)

Strength

Guaranteed to find the shallowest / shortest path.

BFS Numerical Example



Find Goal Node: F

Step 1: Start at 0
Queue: [0], Visited: 0

Step 2: Explore Neighbors
Queue: (Remove 0, Add 1, 2, 3)
Queue: [1, 2, 3], Visited: 0, 1, 2, 3

Step 3: Level 2
Remove 1. Add its neighbors 4 and 5 to the back.
Queue: [2, 3, 4, 5]
Note: Even though we found new nodes (4, 5), the algorithm stays at Layer 1 because 2 and 3 are still at the front of the line.

Step 4: Continue Layer 1
Action: Remove 2. Add its neighbor 6 to the back.
Queue: [3, 4, 5, 6]

Step 5: Goal Check (Reaching Layer 2)
Action: Remove 3. Add its neighbor 7 to the back.
Queue: [4, 5, 6, 7]
Status: Node 7 is now in the queue. BFS will continue processing 4, 5, and 6 before it finally de-queues 7 to complete the search.



| Uninformed Search: DFS

Depth-First Search

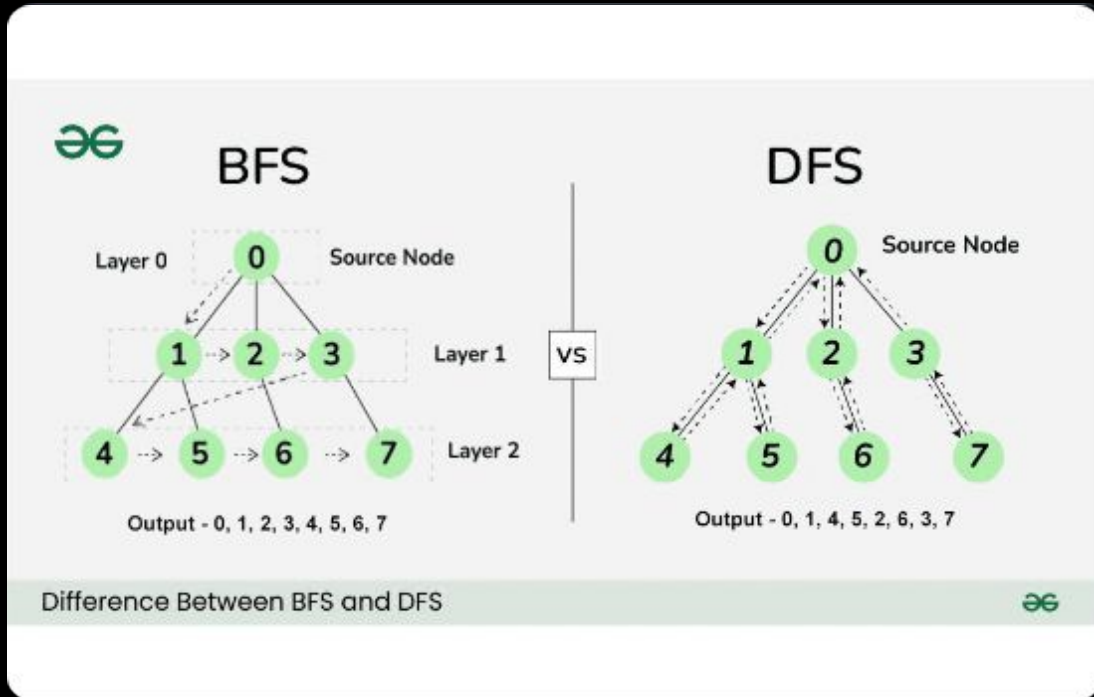
Logic: Follow one path to the very end before backtracking.

≡ Mechanism: **Stack** (Last-In, First-Out)

Strength

Very memory efficient. Deep trees don't require storing every level.

DFS Numerical Example



Find Goal Node: F

Step 1: Start at 0

Stack: [0], Visited: 0

Step 2: Dive Deep (Left Branch)

Pop 0. Push its neighbors 3, 2, 1

Stack: [3, 2, 1], Visited: 0, 1

Step 3: Reach the Bottom (Leaf Node)

Pop 1 (the top of the stack). Push its neighbors 5, 4.

Stack: [3, 2, 5, 4], Visited: 0, 1, 4

Note: Because 4 is now at the top, we visit it immediately, reaching the bottom of the first branch.

Step 4: Backtrack and Explore

Pop 4 (no neighbors to add). Then pop 5.

Stack: [3, 2], Visited: 0, 1, 4, 5

Step 5: Move to the Next Branch

Pop 2. Push its neighbor 6.

Stack: [3, 6], Visited: 0, 1, 4, 5, 2, 6

Step 6: Goal Check

Action: Pop 6. Finally, pop 3 and push its neighbor 7.

Stack: [7]

Visited: 0, 1, 4, 5, 2, 6, 3, 7

Status: Node 7 is reached.



Comparison: BFS vs. DFS

Feature	BFS	DFS
Data Structure	Queue (FIFO)	Stack (LIFO)
Search Pattern	Horizontal (Layer by Layer)	Vertical (Branch by Branch)
Memory	High (stores many nodes per level)	Low (stores nodes along one path)
Final Output	0, 1, 2, 3, 4, 5, 6, 7	0, 1, 4, 5, 2, 6, 3, 7

Scenario: A 3-level binary tree traversal.

BFS Sequence

Root → Level 1 (L, R) → Level 2 (LL, LR...)

Explores wide before going deep.

DFS Sequence

Root → Left → Left-Left → Backtrack →
Left-Right

Can get lost in an infinite branch.

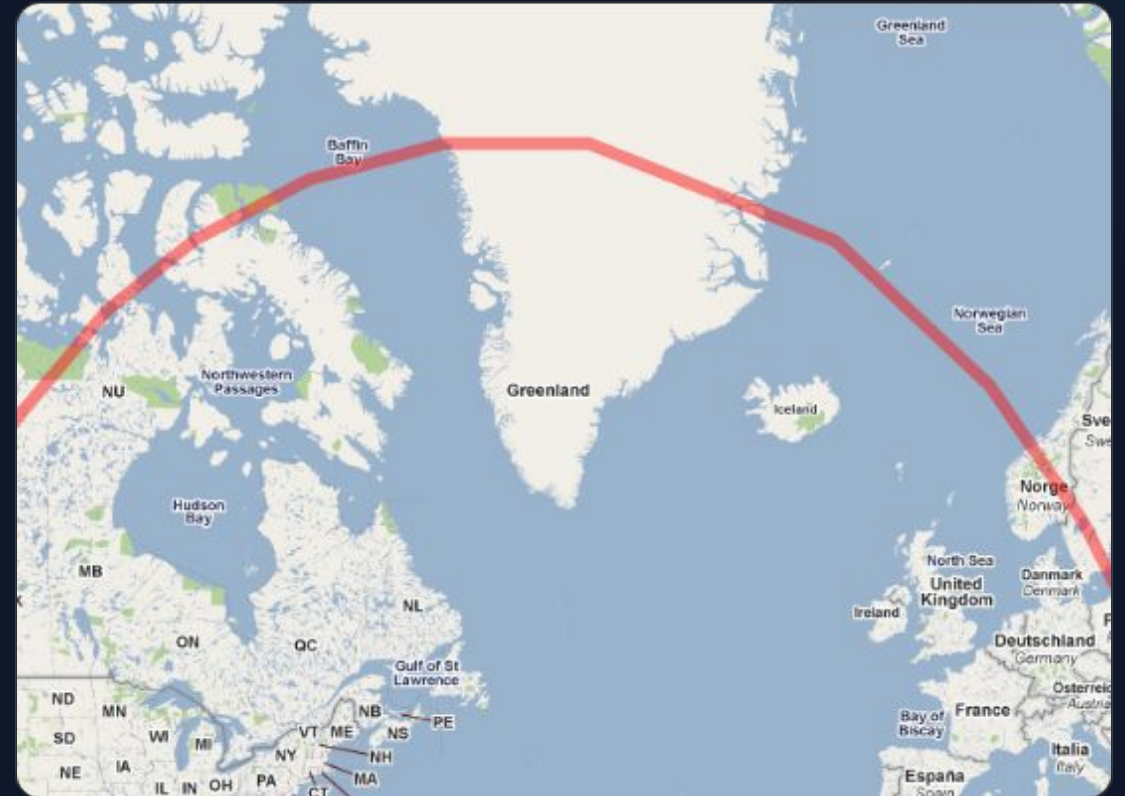
The Concept of Heuristics

Problem: Uninformed search is "blind" (it explores everything).

Solution: Use a Heuristic Function .

$h(n)$

- A "rule of thumb" estimate of how far we are from the goal.
- Example: In navigation, the **straight-line distance** to the target city.



Informed Search: Greedy Best-First



Logic

Always pick the node that looks closest to the goal based **only** on .

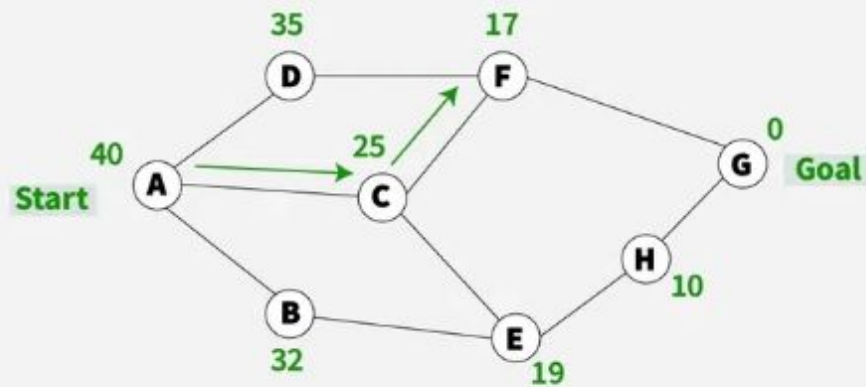
$h(n)$

Risk

It can be led astray by "shortcuts" that lead to dead ends.

Efficiency

Very fast, but **not optimal**.



Best-First Search algorithm



Step 1: Start at Node A ($h=40$)

- **Neighbors:**
 - B ($h=32$)
 - C ($h=25$)
 - D ($h=35$)
- **Action:** Move to **C**, as it has the lowest heuristic value ($h=25$).

Step 2: At Node C ($h=25$)

- **Neighbors:**
 - E ($h=19$)
 - F ($h=17$)
- **Action:** Move to **F**, since it has the lowest heuristic value ($h=17$).

Step 3: At Node F ($h=17$)

- **Neighbor:**
 - G ($h=0$)
- **Action:** Move to **G**, which is the goal node.

Final Path: $A \rightarrow C \rightarrow F \rightarrow G$



| Informed Search: A* Algorithm

$$f(n) = g(n) + h(n)$$



g(n)

Cost so far.
The actual distance
traveled.



h(n)

Estimated cost to
go.
The heuristic.



Balance

Balances "where we've
been" with "where we're
going".

| A* Numerical Example: The Setup

Map Configuration

- Start Node: S
- Goal Node: G

Connections:

- S → A (Cost 2)
- S → B (Cost 5)
- A → G (Cost 10)
- B → G (Cost 4)

Heuristics (Estimates)

Node	h(n) Value
S	8
A	8
B	2
G	0

A* Numerical Example: Step-by-Step

Step 1: At S

Calculate $f(n)$ for neighbors.

Check A:

$$2(g) + 8(h) = 10$$

Check B:

$$5(g) + 2(h) = 7$$

Action: Move to B ($7 < 10$)

Step 2: At B

Calculate path to goal.

Check G:

$$(5+4)(g) + 0(h) = 9$$

Final Path: $S \rightarrow B \rightarrow G$

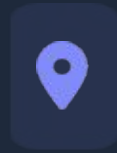
Total Cost: 9

| Search in Modern AI



Game AI

AlphaGo uses "Monte Carlo Tree Search" (MCTS).



Pathfinding

Every GPS app uses a version of A* to find your commute.



LLM Reasoning

Models like OpenAI's "o1" use search-over-thought-chains.

Summary: Search Algorithms

Algorithm	Knowledge	Optimal?	Complete?	Memory
BFS	None	Yes	Yes	High
DFS	None	No	No (if infinite)	Low
Greedy	$h(n)$	No	Yes	Moderate
A*	$g(n)+h(n)$	Yes	Yes	Moderate

Ethical Considerations



Bias

AI inherits human biases from historical data sets.



Displacement

The transition from human labor to ANI/AGI agents.



Alignment

Ensuring Super AI goals match human values.

Self-Assessment

Q1: Which innovation allowed training deep hidden layers in the 1980s?

→ Backpropagation

Q2: If $g(n)=10$ and $h(n)=5$, what is the A^* value?

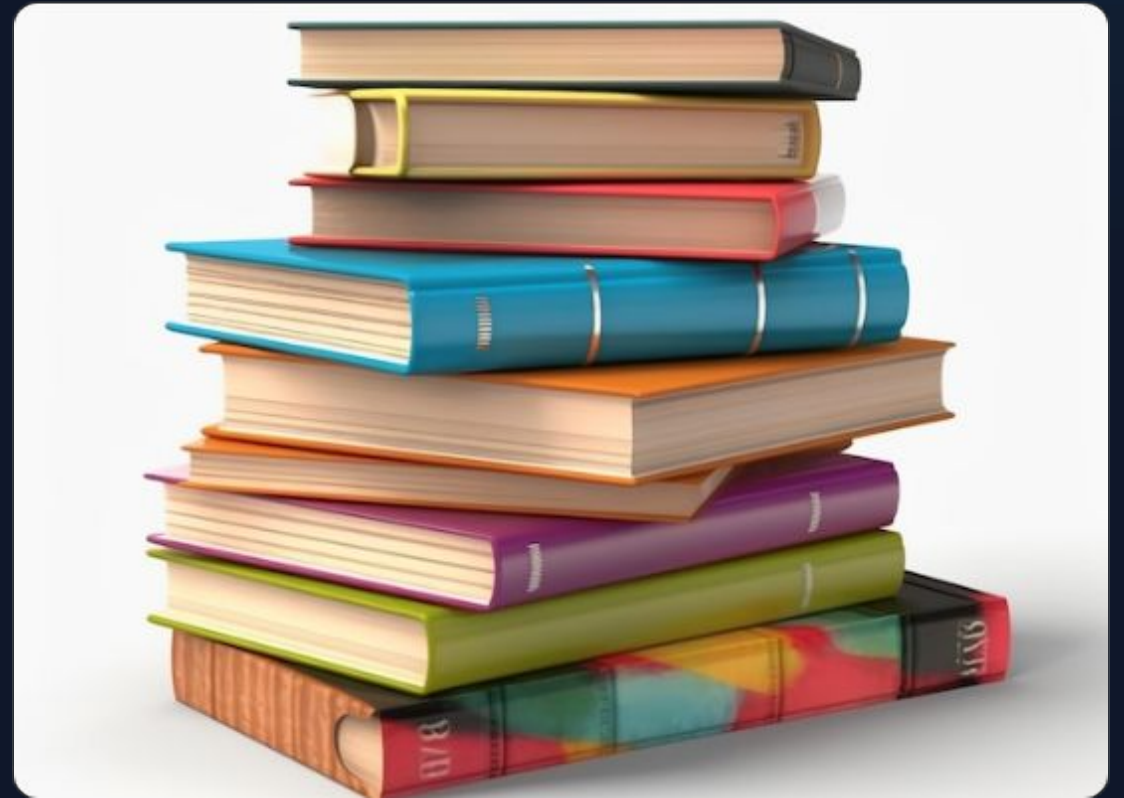
→ 15

Q3: Name one real-world use of CNNs.

→ Medical Imaging / Face ID

References & Further Reading

- **Machine Learning**
Tom Mitchell (Ch. 1)
- **Deep Learning**
Ian Goodfellow (Ch. 1: Historical Trends)
- **Introduction to ML with Python**
Müller & Guido (Ch. 1)



| Image Sources



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